# Previous Year Questions

## Mains (SETS,Relations)

2026

Q1: Let R be a relation defined on the set $\{1,2,3,4\}\times\{1,2,3,4\}$ by $R=\{((a,b),(c,d)):2a+3b=3c+4d\}$. Then the number of elements in R is

(a) 6 (b) 15

(c) 12 (d) 18

Q2: Consider two Sets $A=\{x\in Z:|(|x-3|-3)|\le 1\}$ and $B=\left\{x\in\mathbb{R}-\{1,2\}:\frac{(x-2)(x-4)}{x-1}\log_e(|x-2|)=0\right\}$. Then the number of onto functions $f:A\to B$ is equal to :

(a) 32 (b) 81

(c) 79 (d) 62

Q3: Let $A=\{0,1,2,.....,9\}$. Let R be a relation on A defined by $(x,y)\in R$ if and only if $|x-y|$ is a multiple of 3. Given below are two statements

Statement I : $n(R)=36$

Statement II : R is an equivalence relation

In the light of the above statements, choose the ***correct*** answer from the options given below :

(a) Statement I is correct but Statement II is incorrect.
(b) Both Statement I and Statement II are correct.
(c) Both Statement I and Statement II are incorrect.
(d) Statement I is incorrect but Statement II is correct.

Q4: Let $A=\{-2,-1,0,1,2,3,4\}$. Let R be a relation on A defined by $xRy$ if and only if $2x+y\le 2$. Let $l$ be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l+m+n$ is equal to :

(a) 34 (b) 32

(c) 33 (d) 35

Q5: The number of elements in the relation $R=\{(x,y):4x^2+y^2<52, x,y\in\mathbb{Z}\}$

(a) 86 (b) 67

(c) 89 (d) 77

Q6: Let the relation R on the set $M=\{1,2,3,.....,16\}$ be given by $R=\{(x,y):4y=5x-3, x,y\in M\}$. Then the minimum number of elements required to be added in R, in order to make the relation symmetric , is equal to

(a) 4 (b) 3

(c) 1 (d) 2

Q7: Let $A=\{x:|x^2-10|\le 6\}$ and $B=\{x:|x-2|>1\}$. Then

(a) $A\cup B=(-\infty,1]\cup(2,\infty)$ (b) $B-A=(-\infty,-4)\cup(-2,1)\cup(4,\infty)$

(c) $A-B=[2,3)$ (d) $A\cap B=[-4,-2]\cup[3,4]$

Q8: Let $A=\{2,3,5,7,9\}$. Let R be the relation on A defined by $xRy$ if and only if $2x\le 3y$. Let $l$ be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l+m$ is equal to :

(a) 21 (b) 25

(c) 23 (d) 27

Q9: The number of relations, defined on the set {a,b,c,d}, which are both reflexive and symmetric, is equal to :

(a) 16 (b) 64

(c) 256 (d) 1024

2025

Q1: Let **A** = {0, 1, 2, 3, 4, 5}. Let **R** be a relation on **A** defined by (x, y) ∈ **R** if and only if max{x, y} ∈ {3, 4}. Then among the statements
($S_1$): The number of elements in **R** is 18, and
($S_2$): The relation **R** is symmetric but neither reflexive nor transitive

(a) both are false
(b) only ($S_1$) is true
(c) only ($S_2$) is true
(d) both are true

Q2: Let $A=\{(\alpha,\beta)\in\mathbb{R}\times\mathbb{R}:|\alpha-1|\le 4 \text{ and } |\beta-5|\le 6\}$

$B=\{(\alpha,\beta)\in\mathbb{R}\times\mathbb{R}:16(\alpha-2)^2+9(\beta-6)^2\le 144\}$.

Then

(a) $A\subset B$
(b) $B\subset A$
(c) neither $A\subset B$ nor $B\subset A$
(d) $A\cup B=\{(x,y):-4\le x\le 4,-1\le y\le 11\}$

Q3: Let $A=\{-3,-2,-1,0,1,2,3\}$ and R be a relation on A defined by $xRy$ if and only if $2x-y\in\{0,1\}$. Let $l$ be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+m+n$ is equal to:

(a) 17
(b) 18
(c) 15
(d) 16

Q4: Consider the sets $A=\{(x,y)\in\mathbb{R}\times\mathbb{R}:x^2+y^2=25\}$, $B=\{(x,y)\in\mathbb{R}\times\mathbb{R}:x^2+9y^2=144\}$, $C=\{(x,y)\in\mathbb{Z}\times\mathbb{Z}:x^2+y^2\le 4\}$ and $D=A\cap B$. The total number of one-one functions from the set D to the set C is :

(a) 15120
(b) 18290
(c) 17160
(d) 19320

Q5: Let $A=\{-2,-1,0,1,2,3\}$ . Let R be a relation on $A$ defined by $xRy$ if and only if $y=\max\{x,1\}$ . Let $l$ be the number of elements in $R$ . Let $m$ and $n$ be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+m+n$ is equal to

(a) 11
(b) 12

(c) 14 (d) 13

Q6: Let $A=\{-3,-2,-1,0,1,2,3\}$. Let R be a relation on A defined by $xRy$ if and only if $0 \le x^2+2y \le 4$. Let $l$ be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. Then $l+m$ is equal to:

(a) 18 (b) 20

(c) 17 (d) 19

Q7: Let $A=\{1,2,3,.....,100\}$ and R be a relation on $A$ such that $R=\{(a,b): a=2b+1\}$. Let $(a_1,a_2)$, $(a_2,a_3),(a_3,a_4),.....,(a_k,a_{k+1})$ be a sequence of $k$ elements of R such that the second entry of an ordered pair is equal to the first entry of the next ordered pair. Then the largest integer $k$, for which such a sequence exists, is equal to

(a) 6 (b) 8

(c) 7 (d) 5

Q8: Let $A$ be the set of all functions $f:\mathbb{Z}\to\mathbb{Z}$ and R be a relation on A such that $R=\{(f,g): f(0)=g(1) \quad and \quad f(1)=g(0)\}$ . Then R is :

(a) Symmetric and transitive but not reflexive
(b) Symmetric but neither reflexive nor transitive
(c) Transitive but neither reflexive nor symmetric
(d) Reflexive but neither symmetric nor transitive

Q9: Let $S=\mathbb{N}\cup\{0\}$. Define a relation R from S to $\mathbb{R}$ by:

$R=\left\{(x,y): \log_e y = x\log_e\left(\frac{2}{5}\right), x\in S, y\in\mathbb{R}\right\}$. Then, the sum of all the elements in the range of R is equal to:

(a) $\frac{3}{2}$ (b) $\frac{10}{9}$

(c) $\frac{5}{2}$ (d) $\frac{5}{3}$

Q10: Define a relation R on the interval $\left[0,\frac{\pi}{2}\right)$ by $xRy$ if and only if $\sec^2 x-\tan^2 y=1$. Then R is:

(a) Both reflexive and symmetric but not transitive.
(b) Both reflexive and transitive but not symmetric.

(c) Reflexive but neither symmetric not transitive
(d) An equivalence relation

Q11. The relation $R=\{(x,y): x,y\in\mathbb{Z} \quad and \quad x+y \quad is \quad even\}$ is:

(a) Reflexive and transitive but not symmetric
(b) Reflexive and symmetric but not transitive.
(c) An equivalence relation.
(d) Symmetric and transitive but not reflexive.

Q12: Let $A=\left\{x\in(0,\pi)-\left\{\frac{\pi}{2}\right\}:\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$ and $B=\left\{x\geq 0:\sqrt{x}\left(\sqrt{x}-4\right)-3\left|\sqrt{x}-2\right|+6=0\right\}$. Then $n(A\cup B)$ is equal to

(a) 4 (b) 8

(c) 6 (d) 2

13: Let $X=\mathbb{R}\times\mathbb{R}$. Define a relation R on X as: $(a_1,b_1)R(a_2,b_2)\Leftrightarrow b_1=b_2$

Statement I: R is an equivalence relation.

Statement II: For some $(a,b)\in X$, the set $S=\{(x,y)\in X:(x,y)R(a,b)\}$ represents a line parallel to $y=x$.

In the light of above statements,choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true.
(b) Statement I is true but Statement II is false.
(c) Both Statement I and Statement II are false.
(d) Statement I is false but Statement II is true.

14. Let $A=\{(x,y)\in\mathbb{R}\times\mathbb{R}:|x+y|\geq 3\}$ and $B=\{(x,y)\in\mathbb{R}\times\mathbb{R}:|x|+|y|\leq 3\}$. If $C=\{(x,y)\in A\cap B: x=0 \quad or \quad y=0\}$,then $\sum_{(x,y)\in C}|x+y|$ is :

(a) 18 (b) 24

(c) 15 (d) 12

15. Let $R=\{(1,2),(2,3),(3,3)\}$ be a relation defined on the set $\{1,2,3,4\}$. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:

(a) 9 (b) 8

(c) 7 (d) 10

16. Let $A=\{1,2,3,....,10\}$ and $B=\left\{\frac{m}{n}: m,n\in A, m<n \quad and \quad \gcd(m,n)=1\right\}$. Then $n(B)$ is equal to:

(a) 29 (b) 31

(c) 37 (d) 36

17. The number of non-empty equivalence relations on the set {1,2,3} is:

(a) 7 (b) 4

(c) 5 (d) 6

## 2024

Q1: Let $A=\{2,3,6,8,9,11\}$ and $B=\{1,4,5,10,15\}$. Let R be a relation on $A\times B$ defined by $(a,b)R(c,d)$ if and only if $3ad-7bc$ is an even integer. Then the relation R is

(a) Reflexive but not symmetric.
(b) An equivalence relation.
(c) Reflexive and symmetric but not transitive.
(d) Transitive but not symmetric.

Q2: Let $A=\{1,2,3,4,5\}$. Let R be a relation on A defined by $xRy$ if and only if $4x\le 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A\times A$ that are required to be added to R to make it a symmetric relation. Then m+n is equal to:

(a) 23 (b) 26

(c) 25 (d) 24

Q3: Let $A=\{n\in[100,700]\cap\mathbb{N}$ : n is neither a multiple of 3 nor a multiple of 4} . Then the number of elements in A is

(a) 300 (b) 310

(c) 290 (d) 280

Q4: Let the relations $R_1$ and $R_2$ on the set $X = \{1, 2, 3, ...., 20\}$ be given by $R_1 = \{(x, y) : 2x - 3y = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in $R_1$ and $R_2$, respectively , in order to make the relations symmetric, then M+N equals

(a) 16 (b) 12

(c) 8 (d) 10

Q5: Let a relation R on $\mathbb{N} \times \mathbb{N}$ be defined as : $(x_1, y_1) R (x_2, y_2)$ if and only if $x_1 \le x_2$ or $y_1 \le y_2$. Consider the two statements:

(I) R is reflexive but not symmetric.
(II) R is transitive.

Then which one of the following is true?

(a) Only (II) is correct. (b) Both (I) and (II) are correct.

(c) Neither (I) nor (II) is correct. (d) Only (I) is correct.

Q6: Consider the relations $R_1$ and $R_2$ defined as $aR_1b \Leftrightarrow a^2 + b^2 = 1$ for all $a, b \in \mathbb{R}$ and $(a, b) R_2 (c, d) \Leftrightarrow a + d = b + c$ for all $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$. Then

(a) $R_1$ and $R_2$ both are equivalence relations.
(b) Only $R_1$ is an equivalence relation.
(c) Only $R_2$ is an equivalence relation.
(d) Neither $R_1$ nor $R_2$ is an equivalence relation.

Q7: If R is the smallest equivalence relation on the set $\{1, 2, 3, 4\}$ such that $\{(1, 2)(1, 3)\} \subset R$, then the number of elements in R is

(a) 15 (b) 10

(c) 12 (d) 8

Q8: Let R be a relation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) R (c, d)$ if and only if $ad - bc$ is divisible by 5. Then R is

(a) Reflexive and transitive but not symmetric.
(b) Reflexive and symmetric but not transitive.
(c) Reflexive but neither symmetric nor transitive.
(d) Reflexive, symmetric and transitive.

Q9: Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m,n)$ from the point Q(-2,-3) is:

(a) 8 (b) 10

(c) 4 (d) 6

Q10: Let $S=\{1,2,3,....,10\}$. Suppose M is the set of all the subsets of S, then the relation $R=\{(A,B): A\cap B\neq\phi; A,B\in M\}$ is:

(a) Symmetric only. (b) Reflexive only.

(c) Symmetric and Reflexive only. (d) Symmetric and Transitive only.

## 2023

Q1: Let $A=\{1,3,4,6,9\}$ and $B=\{2,4,5,8,10\}$. Let $R$ be a relation defined on $A\times B$ such that $R=\{((a_1,b_1)(a_2,b_2)): a_1\leq b_2 \quad and \quad b_1\leq a_2\}$. Then the number of elements in the set R is:

(a) 180 (b) 26

(c) 52 (d) 160

Q2: An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events?

(a) 10 (b) 15

(c) 21 (d) 9

Q3: Let $A=\{2,3,4\}$ and $B=\{8,9,12\}$. Then the number of elements in the relation $R=\{((a_1,b_1),(a_2,b_2))\in(A\times B, A\times B): a_1$ divides $b_2$ and $a_2$ divides $b_1\}$ is:

(a) 18 (b) 24

(c) 36 (d) 12

Q4: Let $A=\{1,2,3,4,5,6,7\}$. Then the relation $R=\{(x,y)\in A\times A: x+y=7\}$ is:

(a) Reflexive but neither symmetric nor transitive.
(b) Transitive but neither symmetric nor reflexive.

(c) Symmetric but neither reflexive nor transitive.
(d) An equivalence relation.

Q5: Let $P(S)$ denote the power set of $S=\{1,2,3,....,10\}$. Define the relations $R_1$ and $R_2$ on $P(S)$ as $AR_1B$ if $\left(A\cap B^c\right)\cup\left(B\cap A^c\right)=\phi$ and $AR_2B$ if $A\cup B^c=B\cup A^c, \forall A,B\in P\left(S\right)$. Then :

(a) only $R_2$ is an equivalence relation
(b) both $R_1$ and $R_2$ are not equivalence relations
(c) both $R_1$ and $R_2$ are equivalence relations
(d) only $R_1$ is an equivalence relation

Q6: Let R be a relation on $\mathbb{R}$, given by $R=\left\{\left(a,b\right):3a-3b+\sqrt{7}\right.$ is an irrational number }. Then R is

(a) an equivalence relation
(b) reflexive and symmetric but not trasitive
(c) reflexive and transitive but not symmetric
(d) reflexive but neither symmetric nor transitive

Q7: Among the relations $S=\left\{\left(a,b\right):a,b\in\mathbb{R}-\left\{0\right\},2+\frac{a}{b}>0\right\}$

and $T=\left\{\left(a,b\right):a,b\in\mathbb{R},a^2-b^2\in\mathbb{Z}\right\}$,

(a) S is transitive but T is not
(b) Both S and T are symmetric
(c) Neither S nor T is transitive
(d) T is symmetric but S is not

Q8: Let R be a relation on $\mathbb{N}\times\mathbb{N}$ defined by $\left(a,b\right)R\left(c,d\right)$ if and only if $ad(b-c)=bc(a-d)$. Then R is

(a) Symmetric and transitive but not reflexive
(b) Reflexive and symmetric but not transitive
(c) Trasitive but neither reflexive nor symmetric
(d) Symmetric but neither reflexive nor transitive

Q9: The minimum number of elements that must be added to the relation $R=\left\{\left(a,b\right),\left(b,c\right)\right\}$ on the set $\left\{a,b,c\right\}$ so that it becomes symmetric and transitive is :

(a) 7 (b) 3

(c) 4 (d) 5

Q10: Let R be a relation defined on $\mathbb{N}$ as $aRb$ if $2a+3b$ is a multiple of $5, a, b \in \mathbb{N}$. Then R is

(a) An equivalence relation (b) non-reflexive

(c) Symmetric but non transitive (d) transitive but not symmetric

Q11: The relation $R = \{(a,b) : \gcd(a,b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$ is:

(a) Reflexive but not symmetric (b) Transitive but not reflexive

(c) Symmetric but not transitive (d) neither symmetric nor transitive

## 2022

Q1: Let R be a relation from the set $\{1, 2, 3, \ldots\ldots, 60\}$ to itself such that $R = \{(a,b) : b = pq,$ where $p, q \geq 3$ are prime numbers}. Then, the number of elements in R is :

(a) 600 (b) 660

(c) 540 (d) 720

Q2: For $\alpha \in \mathbb{N}$ , consider a relation R on $\mathbb{N}$ given by $R = \{(x, y) : 3x + \alpha y$ is a multiple of 7}. The relation R is an equivalence relation if and only if :

(a) $\alpha = 14$ (b) $\alpha$ is a multiple of 4

(c) 4 is the remainder when $\alpha$ is divided by 10

(d) 4 is the remainder when $\alpha$ is divided by 7

Q3: Let $R_1$ and $R_2$ be two relations defined on $\mathbb{R}$ by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$ . Then

(a) $R_1$ is an equivalence relation but not $R_2$

(b) $R_2$ is an equivalence relation but not $R_1$

(c) Both $R_1$ and $R_2$ are equivalence relations

(d) Neither $R_1$ nor $R_2$ is an equivalence relation

Q4: Let a set $A = A_1 \cup A_2 \cup \ldots\ldots \cup A_k$ , where $A_i \cap A_j = \varphi$ for $i \neq j, 1 \leq i, j \leq k$ . Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i, 1 \leq i \leq k\}$ . Then R is :

(a) Reflexive , symmetric but not transitive

(b) Reflexive, transitive but not symmetric
(c) Reflexive but not symmetric and transitive
(d) An equivalence relation

Q5: Let $R_1 = \{(a,b) \in \mathbb{N} \times \mathbb{N} : |a-b| \le 13\}$ and $R_2 = \{(a,b) \in \mathbb{N} \times \mathbb{N} : |a-b| \ne 13\}$. Then on N

(a) Both $R_1$ and $R_2$ are equivalence relations
(b) Neither $R_1$ nor $R_2$ is an equivalence relation
(c) $R_1$ is an equivalence relation but $R_2$ is not
(d) $R_2$ is an equivalence relation but $R_1$ is not

## 2021

Q1: Which of the following is not correct for relation R on the set of real numbers ?

(a) $(x,y) \in R \Leftrightarrow 0 \le |x| - |y| \le 1$ is neither transitive nor symmetric.
(b) $(x,y) \in R \Leftrightarrow 0 < |x-y| \le 1$ is symmetric and transitive.
(c) $(x,y) \in R \Leftrightarrow 0 \le |x| - |y| \le 1$ is reflexive but not symmetric.
(d) $(x,y) \in R \Leftrightarrow |x-y| \le 1$ is reflexive and symmetric.

Q2: Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set :

(a) $\{80, 83, 86, 89\}$
(b) $\{84, 86, 88, 90\}$
(c) $\{79, 81, 83, 85\}$
(d) $\{84, 87, 90, 93\}$

Q3: Let $\mathbb{N}$ be the set of natural numbers and a relation R on $\mathbb{N}$ be defined by $R = \{(x,y) \in \mathbb{N} \times \mathbb{N} : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is:

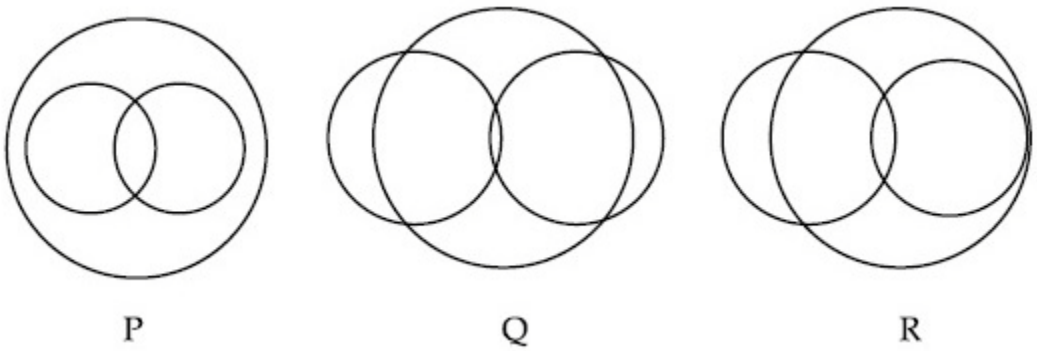
(a) Symmetric but neither reflexive nor transitive
(b) Reflexive but neither symmetric nor transitive
(c) Reflexive and symmetric but not transitive
(d) An equivalence relation

Q4: Define a relation R over a class of $n \times n$ real matrices A and B as "ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$". Then which of the following is true?

(a) R is reflexive, transitive but not symmetric

(b) R is symmetric, transitive but not reflexive.
(c) R is reflexive, symmetric but not transitive.
(d) R is an equivalence relation.

Q5: In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?




(a) Q and R
(b) None of these
(c) P and R
(d) P and Q

Q6: Let A = {2, 3, 4, 5, ....., 30} and '$\simeq$' be an equivalence relation on $A \times A$ , defined by (a, b) $\simeq$ (c, d), if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to :

(a) 5
(b) 6
(c) 8
(d) 7

Q7: The number of elements in the set $\{x \in R : (|x| - 3)|x + 4| = 6\}$ is equal to:

(a) 4
(b) 2
(c) 3
(d) 1

Q8: Let R = {(P, Q) | P and Q are at the same distance from the origin} be a relation, then the equivalence class of (1, -1) is the set :

(a) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$
(b) $S = \{(x, y) \mid x^2 + y^2 = 2\}$
(c) $S = \{(x, y) \mid x^2 + y^2 = 1\}$
(d) $S = \{(x, y) \mid x^2 + y^2 = 4\}$

# 2020

Q1 : Consider the two sets : $A = \{m \in \mathbb{R}$ : both the roots of $x^2 - (m+1)x + m + 4 = 0$ are real} and $B = [-3, 5)$. Which of the following is not true?

(a) $A \cup B = (-\infty, -3) \cup (5, \infty)$

(b) $A \cap B = \{-3\}$

(c) $B - A = (-3, 5)$

(d) $A \cup B = \mathbb{R}$

Q2 : Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$, where each $X_i$ contains 10 elements and each $Y_i$ contains 5 elements. If each element of the set T is an element of exactly 20 of sets $X_i$'s and exactly 6 of sets $Y_i$'s then n is equal to

(a) 15

(b) 30

(c) 50

(d) 45

Q3 : A survey shows that 73% of the persons working in an office like coffee, whereas 65% like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be :

(a) 63

(b) 54

(c) 38

(d) 36

Q4 : Let $A = \{x : |x| < 2\}$ and $b = \{x : |x - 2| \geq 3\}$ then :

(a) $A \cap B = [-2, -1]$

(b) $A \cup B = \mathbb{R} - (2, 5)$

(c) $A - B = [-1, 2]$

(d) $B - A = \mathbb{R} - (-2, 5)$

Q5: A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x% of the people read both the newspapers, then a possible value of x can be:

(a) 37

(b) 65

(c) 29

(d) 55

Q6: Let $R_1$ and $R_2$ be two relation defined as follows :
$R_1 = \{(a, b) \in R^2 : a^2 + b^2 \in Q\}$ and
$R_2 = \{(a, b) \in R^2 : a^2 + b^2 \notin Q\}$,
where Q is the set of all rational numbers. Then :

(a) Neither $R_1$ nor $R_2$ is transitive.

(b) $R_2$ is transitive but $R_1$ is not transitive

(c) $R_1$ and $R_2$ are both transitive

(d) $R_1$ is transitive but $R_2$ is not transitive

Q7: If R = {(x, y) : x, y $\in$ Z, $x^2 + 3y^2 \leq 8$} is a relation on the set of integers Z, then the domain of $R^{-1}$ is :

(a) {0,1} (b) {-2,-1,1,2}

(c) {-1,0,1} (d) {-2,-1,0,1,2}

## 2019

Q1 : Let A, B and C be sets such that $\varphi \neq A \cap B \subseteq C$. Then which of the following statements are not true?

(a) If $(A-C) \subseteq B$, then $A \subseteq B$ (b) If $(A-B) \subseteq C$ ,then $A \subseteq C$

(c) $(C \cup A) \cap (C \cup B) = C$ (d) $B \cap C \neq \phi$

Q2: Let $S = \{1, 2, 3, ....., 100\}$. The number of non-empty subsets A of S such that the product of elements in A is even is :

(a) $2^{100} - 1$ (b) $2^{50}(2^{50} - 1)$

(c) $2^{50} - 1$ (d) $2^{50} + 1$

Q3: Let Z be the set of integers.

If $A = \{x \in Z : 2^{(x+2)}(x^2 - 5x + 6) = 1\}$ and

$B = \{x \in Z : -3 < 2x - 1 < 9\}$, then the number of subsets of the set $A \times B$, is :

(a) $2^{15}$ (b) $2^{18}$

(c) $2^{12}$ (d) $2^{10}$

Q4: Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-

(a) 13.5 (b) 13

(c) 12.8 (d) 13.9

Q5: In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is

(a) 42 (b) 102

(c) 1 (d) 38

## 2018

Q1: Let **N** denote the set of all natural numbers. Define two binary relations on **N** as R = {(x, y) $\in$ **N** $\times$ **N** : 2x + y = 10} and $R_2$ = {(x, y) $\in$ **N** $\times$ **N** : x + 2y = 10}. Then :

(a) Range of $R_1$ is {2,4,8}. (b) Range of $R_2$ is {1,2,3,4}.

(c) Both $R_1$ and $R_2$ are symmetric relations

(d) Both $R_1$ and $R_2$ are transitive relations

Q2: Two sets A and B are as under:

$$A = \{(a,b) \in \mathbb{R} \times \mathbb{R} : |a-5| < 1 \quad and \quad |b-5| < 1\};$$

$$B = \{(a,b) \in \mathbb{R} \times \mathbb{R} : 4(a-6)^2 + 9(b-5)^2 \le 36\};$$

Then

(a) Neither $A \subset B$ nor $B \subset A$ (b) $B \subset A$

(c) $A \subset B$ (d) $A \cap B = \varphi$ (an empty set)

Q3: Consider the following two binary relations on the set A = {a, b, c} :
$R_1$ = {(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)} and
$R_2$ = {(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)}.
Then

(a) Both $R_1$ and $R_2$ are not symmetric.

(b) $R_1$ is not symmetric but it is transitive.

(c) $R_2$ is symmetric but it is not transitive.

(d) Both $R_1$ and $R_2$ are transitive.

## 2016

Q1: Let $P = \left\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\right\}$ $Q = \left\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\right\}$ be two seats. Then

(a) $P \subset Q$ and $Q - P \neq \phi$

(b) $Q \not\subset P$

(c) $P \not\subset Q$

(d) $P = Q$

## 2015

Q1: Let A and B be two sets containing four and two elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is:

(1) 256

(2) 275

(3) 510

(4) 219

## 2014

1. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \left\{9(n-1) : n \in N\right\}$, where N is the set of natural numbers, then $X \cup Y$ is equal to

(1) N

(2) Y – X

(3) X

(4) Y

## 2013

1. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is

(1) 220

(2) 219

(3) 211

(4) 256

## 2010

1. Let S be a non-empty subset of R. Consider the following statement:

P: There is a rational number $x \in S$ such that x > 0.

Which of the following statements is the negation of the statement P?

(1) There is no rational number $x \in S$ such that $x \leq 0$
(2) Every rational number $x \in S$ satisfies $x \leq 0$
(3) $x \in S$ and $x \leq 0 \Rightarrow x$ is not rational
(4) There is a rational number $x \in S$ such that $x \leq 0$

2. Consider the following relations:

R = {(x,y)|x,y are real numbers and x=wy for some rational number w};

S = $\left\{\left(\frac{m}{n}, \frac{p}{q}\right) | \, m, n, p \, and \, q \, are \, \text{integers} \, such \, n, q \neq 0 \, and \, qm = pn\right\}$. Then

(1) neither R nor S is an equivalence relation
(2) S is an equivalence relation but R is not an equivalence relation
(3) R and S both are equivalence relations
(4) R is an equivalence relation but S is not an equivalence relation

## 2009

1. If A, B nd C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then

(1) A = B
(2) A = C
(3) B = C
(4) $A \cap B = \phi$

## 2008

1. Let R be the real line. Consider the following subsets of the plane $R \times R$.

S = {(x,y): y = x + 1 and 0 < x <2 }, T = {(x,y):x – y is an integer}. Which one of the following is true?

(1) neither S nor T is an equivalence relation on R
(2) Both S and T are equivalence relations on R
(3) S is an equivalence relation on R but T is not
(4) T is an equivalence relation on R but S is not

## 2006

1. Let W denote the words in the English dictionary. Define the relation R by :

R = {(x,y)ε$W \times W$ | the words x and y have at least one letter in common}. Then R is

(1) Not reflexive, symmetric and transitive (2) reflexive, symmetric and not transitive

(3) reflexive, symmetric and transitive (4) reflexive, not symmetric and transitive

## 2005

1. Let R = {(3,3), (6,6), (9,9),(12,12),(6,12),(3,9),(3,12),(3,6)} be a relation on the set A = {3,6,9,12} be a relation on the set A = {3,6,9,12}. The relation is

(1) reflexive and transitive only (2) reflexive only

(3) an equivalence relation (4) reflexive and symmetric only

## 2004

1. Let R = {(1,3),(4,2),(2,4),(2,3),(3,1)} be a relation on the set A = {1,2,3,4}. The relation R is

(1) a function (2) reflexive

(3) not symmetric (4) transitive